Session 5

1. Create two tables in SQL: influencers (with influencer_id as PRIMARY KEY and name) and posts (with post_id as PRIMARY KEY, influencer_id as FOREIGN KEY, and caption). Insert at least 3 influencers and 2 posts for each influencer.

➔

```
USE instagram;

CREATE TABLE influencers(
   influencers_id INT(11) AUTO_INCREMENT PRIMARY KEY,
   name VARCHAR(255)
);

CREATE TABLE posts(
   post_id INT(11) PRIMARY KEY AUTO_INCREMENT,
   influencers_id INT(11),
   caption VARCHAR(255),
   FOREIGN KEY (influencers_id) REFERENCES influencers(influencers_id)
);

INSERT INTO influencers(name)
VALUES
('Jay Meghani'),
('Mitansh Raval'),
('Sahdev Mali');

INSERT INTO posts(influencers_id,caption)
VALUES
(1,'Trek to Manali'),
(1,'Kauri Pass Views'),
(2,'Java Developers'),
(3,'Mathematician Estudent'),
(3,'Student of Ramanujan');
```

Output | Result #6

influencers_id	name
1	Jay Meghani
2	Mitansh Raval
3	Sahdev Mali

Output | Result #6

post_id	influencers_id	caption
1	1	Trek to Manali
2	1	Kauri Pass Views
3	2	Java Developers
4	3	Mathematician Estudent
5	3	Student of Ramanujan

2. Write an SQL query using INNER JOIN to display each post's caption along with the name of the influencer who posted it, based on the tables you created.

➔

```
USE instagram;

SELECT posts.caption, influencers.name
FROM posts
INNER JOIN influencers
ON posts.influencers_id = influencers.influencers_id;
```

Output | Result #2

caption	name
Trek to Manali	Jay Meghani
Kauri Pass Views	Jay Meghani
Java Developers	Mitansh Raval
Mathematician Estudent	Sahdev Mali
Student of Ramanujan	Sahdev Mali

3. Write an SQL query using LEFT JOIN to show all influencers and their posts, including influencers who haven't posted anything yet. If an influencer has no posts, display 'No Posts' in the caption column. Hint: Use IFNULL or COALESCE to handle NULL values in the result.

➔

```
SELECT influencers.name,
    IFNULL(post.caption, 'No Posts') AS caption
FROM influencers
LEFT JOIN posts
ON influencers.influencers_id = posts.influencers_id;
```

Output | Result #2

name	caption
Jay Meghani	Trek to Manali
Jay Meghani	Kauri Pass Views
Mitansh Raval	Java Developers
Sahdev Mali	Mathematician Estudent
Sahdev Mali	Student of Ramanujan

4. Write an SQL query using RIGHT JOIN to list all posts and the corresponding influencer's name, ensuring that even posts without a matching influencer_id (if any) are shown.

➔

```
USE instagram;

SELECT posts.caption, influencers.name
FROM influencers
RIGHT JOIN posts
ON influencers.influencers_id = posts.influencers_id;
```

Output | Result #2

caption	name
Trek to Manali	Jay Meghani
Kauri Pass Views	Jay Meghani
Java Developers	Mitansh Raval
Mathematician Estudent	Sahdev Mali
Student of Ramanujan	Sahdev Mali